

Emily Barosin, EIT

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EDUCATION AND AWARDS

Dartmouth College — Hanover, NH.

June 2025

B.E., Environmental Engineering + B.A., Engineering Sciences, cum laude

- GPA: 3.85 / 4.0
- High Honors Thesis: A structural health monitoring system for underwater turbines using passive acoustic monitoring techniques
- Cheer Team (Captain): Four years, 20+ hours per week; raised ~\$30,000 USD for the program during tenure

TECHNICAL SKILLS

Computational / Software: Python (PyTorch, TensorFlow), MATLAB, Julia, acoustic modeling, CFD (Ansys)

Scientific: Signal processing, acoustics research, acoustic vector sensors, machine learning, numerical methods

RESEARCH AND WORK EXPERIENCE

Integral Consulting, Inc.

April 2024 – Present

Acoustics Scientist — Santa Cruz, CA

- Invented novel methods to process acoustic vector sensor data, producing a patent, two peer-reviewed publications, and three conference presentations, including an invited talk at the Acoustical Society of America (May 2026).
- Leading technical development and implementation of a navigation-aided sonar perception model on a UUV, that fuses vehicle control telemetry with sonar streams to classify features of interest in floating offshore wind lease areas.
- Architecting a low-cost, small, and efficient passive acoustic monitoring system to detect vessel activity in marine protected areas, with custom logging algorithms and COTS components. Completed initial tests near South American fishery this year.
- Pioneering new machine-learning-based post-processing techniques of hydrostatic wave models in XBeach for surf quality analysis
- Regularly deliver technical briefings to clients, translating analysis into actionable recommendations and supporting proposal development

Dartmouth College, Thayer School of Engineering

April 2024 – June 2025

Undergraduate Thesis Student, Mazilu Fellow — Hanover, NH

Advisor: Benoit Cushman-Roisin, PhD

- Designed and executed a controlled tank experiment to detect structural damage in underwater turbine blades via passive acoustics; benchmarked random forest and CNN classifiers across a sweep of acoustic features (MFCCs, spectral shape descriptors, etc.) to identify the optimal model-feature pairing for discriminating intact vs. damaged blade signatures.
- Engineered an end-to-end real-time fault diagnosis pipeline from hydrophone front-end through edge inference, enabling pre-failure detection of structural faults in deployment scenarios.

Dartmouth College, Thayer School of Engineering

April 2024 – June 2025

MECC Competition Team Lead, Engineering Capstone Student — Hanover, NH

- Invented and fabricated a power-productive vertical-axis tidal turbine for use in low-flow-speed environments
- Created and coordinated Phase I testing protocol at UNH Chase Ocean Engineering Laboratory
- Evaluated turbine performance and structural health from testing data and CFD modeling in Ansys Fluent
- Represented the engineering team in meetings with industry stakeholders and MECC administrators

Stony Brook University, School of Marine and Atmospheric Sciences

June 2022 – Aug 2023

Research Assistant — Southampton, NY

Supervisor: Joe Warren

- Developed and conducted a field experiment to investigate the diurnal effects on sound propagation in eelgrass meadows in shallow estuarine environments
- Analyzed data for propagation characterization and compared to model predictions

SELECTED CONFERENCE PRESENTATIONS

Barosin, E.B., Oyanedel, R., Raghukumar, K., Spada, F., (2026, *accepted*). *Demonstration of a Real-Time, Scalable, Acoustic Monitoring System for IUU Fishing Detection in Marine Protected Areas*. Oceans 2026. Monterey, CA, October 2026

Raghukumar K., **Barosin, E.B.**, Chang, G., O'Brien, J., Voorakkara, A., Kramer, S., Escajeda, E., Chamberlain, P., (2026). *Development of an autonomous detection and mitigation framework for entanglements around offshore wind farms*. State of the Science on Offshore Energy, Wildlife, and Fisheries. Long Island, NY, June 2026.

Barosin, E. B., & Raghukumar, K. (2026). *Directional acoustic filtering for sound field estimation and verification*. State of the Science on Offshore Energy, Wildlife, and Fisheries. Long Island, NY, June 2026.

Barosin, E. B., Raghukumar, K., Chamberlain, P., Chang, G., Chao, Y., Griffin, S., Joseph, J., Mazzloff, M., & Ryan, J. (2026, *invited*). *Demonstration of a directional acoustic profiling float powered by ocean thermal energy conversion*. Acoustical Society of America. Philadelphia, PA, May 2026.

Raghukumar, K., **Barosin, E. B.**, Cotter, E. D., Haxel J., Polagye, B., Bassett, C., (2025). *Directional acoustic measurements around a small-scale tidal turbine*. Acoustical Society of America. New Orleans, LA, May 2025.

Barosin, E. B., & Raghukumar, K. (2025). *Towards more accurate sound field verification using directional acoustic filtering*. Effects of Noise on Aquatic Life. Prague, Czech Republic, June 2025.

Barosin, E. B., & Raghukumar, K. (2025). *Towards more accurate sound field verification using directional acoustic filtering*. Acoustical Society of America. New Orleans, LA, May 2025.

Barosin, E. B., & Warren, J. (2023). *Diurnal measurements of sound attenuation (100 Hz–10 kHz) in a shallow seagrass (*Zostera marina*) meadow are impacted by photosynthetic activity and water column height*. Acoustical Society of America. Sydney, Australia, December 2023.

PUBLICATIONS

Barosin, E. B., & Raghukumar, K. (2026, *accepted*). On directional filtering for sound source verification and source level estimation. In A. N. Popper, J. Sisneros, P. Lepper, & K. Vigness-Raposa (Eds.), *The Effects of Noise on Aquatic Life IV*. Springer, Cham.

Barosin, E. B., & Raghukumar, K. (2025). Towards more accurate sound field verification using directional acoustic filtering. *JASA Express Letters*, 5(4). <https://doi.org/10.1121/10.0036394>

LEADERSHIP AND SCIENCE COMMUNICATION

Irving Institute Ambassador, Dartmouth College 2022 – 2024
Facilitated outreach events to energy industry stakeholders, students, and alumni. Communicated the Institute's goals and projects to audiences with a diverse array of technical backgrounds.

Dartmouth Humanitarian Engineering Project Co-Lead, Dartmouth College 2022 – 2024
Helped design and fabricate a gravity-powered washing machine for communities with unreliable access to electricity. Represented the engineering team to the Thayer School administration and the target-impact group. Organized technical training of new members, delegated project tasks, and served as environmental tech lead.

RESEARCH GRANTS AND AWARDS

Mazilu Fellow — Thayer School of Engineering, Dartmouth College \$10,000 • 2024–2025

Marine Energy Collegiate Competition Award \$20,000 • 2024–2025

— Dept. of Energy, Water Power Technologies Office

OOI Summer Acoustics Program — National Science Foundation	\$3,000 • 2025
Independent Student Researcher Grant (three projects funded) — Irving Institute for Energy and Society, Dartmouth College	\$23,000 • 2023
Down Under Funder — Acoustical Society of America	\$1,000 • 2023
SELF Grant — Dartmouth College Center for Professional Development	\$1,500 • 2022